

tractor. Horses were the technology for transportation, and there was so much technology which went into the horse—somebody had to invent the saddle, the spur, and the horseshoe—all of which were connected with the horse to really make it into a transportation machine. There has been the automotive revolution and we will at some point see the autonomy revolution as well.

Automation is not always about labour-saving. Think about welding robots. Today, almost all welding for making the car is done by robotic arms. Initially, some people did it, but you can't manufacture many cars if you weld manually. It's beyond human capability to weld with that accuracy repeatedly, at that speed. So, the only option is to have a robotic arm which does it. One could argue why there is a conveyor belt when somebody could lift the packet and run. The person will never be able to meet the speed of the conveyor moving something. Similarly, sometimes a robot is required because conveyors are not practical or you have large material that humans cannot carry. You will not get the repeatability, accuracy, and reliability with humans that you get with robots, especially in the factory.

When you come to public areas, or deliveries, the challenge is of availability now. There aren't enough young people who want to do this work, especially in the West. So it's just not viable anymore. Even in India, there aren't enough people available in the country to work in factories. The total manufacturing workforce is 50 million people typically, in India right now. We produce only 2.5 million people annually from ITIs. That is hardly 5%. So we'd just be replenishing the attrition whereas this 50 million should become 100 million over time. To increase this figure we need to increase our manufacturing output, which is not going to happen unless we add enough skilled people to that pool. The scale of operations is also increas-

ing. We can't reach that scale at all without automation. There's just no way. On the assembly line, you have a car coming out every 30 seconds or one minute. A full car! How do you do that if you have not automated a lot of stuff!

So what are the challenges that you faced in building this product?

For any startup, life is only full of challenges. It's difficult to name very few things. Since we're doing multidisciplinary engineering we do everything. We make the mechanical vehicle, handle the electronics, and cover the software stack as well. So any multidisciplinary engineering system, by definition, becomes a little challenging. Apart from that, supply-chain challenges have been hard for us, like for many people in the industry. These challenges exist in any startup's journey, especially when it comes to hardware startups. Running a product in the lab is reasonably easy, but taking it to a customer's production site is very challenging. Crossing the chasm from lab to production is difficult. Another challenge lies in increasing the production capacity, while ensuring the same quality repeatedly. At different stages, one has to keep crossing these challenges, and they never stop. Customers will keep coming in. We have to support our old products while developing new products. Product management is also very complex and challenging, especially with hardware. You can just update the software in so many places, but supporting old hardware is not that easy. We have to do it because once a customer buys a product, it is with the expectation that we will support them for the next 5-10 years, and we have to live by that commitment.

What are the future trends and technologies that you expect to see in this space?

Firstly, we are still in the very early days of this space. You'll see all the line-following robots in many of the factories. But the penetration of these modern, without-line, flexible, mobile robots is still very low. It's minuscule. I would share another interesting statistic. There is a number which people track which is called robot density, which calculates the number of robots per 10,000 workers. The world average of that number is around 126. The US is at 250. South Korea is in the 950 range, which is a benchmark even for the US which has a long way to go. China has gone from 50 to 250 in just five years. India is still at 4. There is so much that we need to do to catch up not just with the world, but just with our neighbours. So, I would say forget future trends, let's catch up with the present first. We have to do that at double the speed. We have to compete globally as we are trying to attract global manufacturing into India. We cannot do that if we are not at the level of our global competitors. Such automation is important to produce things at the same quality scale and cost.

How do you see the global competition in this space?

We are quite unique globally also in our space. The approach of self-driving car like technology inside factories is still novel. There are demos but significant production deployment as we have done across more than 25 locations is rare. People have tried it in other use cases. Outrider will move full trucks to the yard of the warehouse or the factory. Einride is doing teleoperated automation where somebody is remotely driving; they're not trying to do full autonomy. Different people have different approaches and everybody has some market because there are so many different things that customers want. No one company will do everything for them. I would like to see more companies in India doing this. **EFY**